

Principal Components Analysis (PCA): Linear Algebra Background

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Homogeneous Linear Equations

- $A \iff B$: *An integer is even (A) if and only if an integer is divisible by 2 (B).*
- $\mathbf{Ax} = \mathbf{0}$ has a non-trivial solution if and only if columns of $\mathbf{A}$ are linearly dependent.
- $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution if and only if columns of $\mathbf{A}$ are linearly independent.
- Does $\mathbf{A}$ need to be a square matrix ?

$$\mathbf{Ax} = \mathbf{0}$$

$$\Rightarrow [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Rightarrow x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{0}$$

Determinants

- Under row operations, a zero determinant cannot become non-zero and vice-versa.
- Columns of $\mathbf{A}$ linearly independent: By definition of linear independence, the only possible solution to $\mathbf{Ax} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$ i.e. $I_{n \times n} \mathbf{x} = \mathbf{0}$. $\Rightarrow$ The determinant of $I_{n \times n}$ is non-zero $\Rightarrow$ The determinant of $\mathbf{A}$ is non-zero and cannot be zero under any circumstances.
- Columns of $\mathbf{A}$ are linearly dependent $\Rightarrow$ One row is a linear combination of some of the other rows (row rank = column rank) $\Rightarrow$ Row operations can create a row of zeros.

Homogeneous Linear Equations: Square Matrices

- Consider a square matrix $\mathbf{A}$.
- We denote the determinant of a square matrix $\mathbf{A}$ by $|\mathbf{A}|$.
- $|\mathbf{A}| = 0$ if and only if $\mathbf{A}$ has linearly dependent columns.
- $|\mathbf{A}| \neq 0$ if and only if $\mathbf{A}$ has linearly independent columns.
- The system of linear equations $\mathbf{Ax} = \mathbf{0}$ has non-trivial solutions if and only if $|\mathbf{A}| = 0$.

Rank-Nullity Theorem

- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The *nullity* of $\mathbf{A}$ is the dimension of its null space, that is, $\text{nullity}(\mathbf{A}) = \dim\{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0}\}$.
- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a matrix. Then $\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$.

Eigenvalues and Eigenvectors

- Suppose $\mathbf{Ax} = \lambda\mathbf{x}$. Here, $\mathbf{x}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ provided $\mathbf{x} \neq \mathbf{0}$. λ can equal 0.
- The relation $\mathbf{Ax} = \lambda\mathbf{x}$ requires $\mathbf{A}$ to be a square matrix. $\mathbf{A}$ is $n \times n$ when $\mathbf{x}$ is $n \times 1$.
- $\mathbf{Ax} = \lambda\mathbf{x} \Rightarrow (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$
- The determinant $|\mathbf{A} - \lambda\mathbf{I}|$ is the characteristic polynomial of the matrix $\mathbf{A}$.
- For a non-trivial solution $\mathbf{x}$ to exist for $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$, we must have $|\mathbf{A} - \lambda\mathbf{I}| = 0$.
- Eigenvalues are the roots of the characteristic polynomial.

Geometric and Algebraic multiplicity

- Let $\mathbf{A} \in \mathbb{F}^{n \times n}$ and let λ be an eigenvalue of $\mathbf{A}$. The *eigenspace* corresponding to λ is $E_\lambda = \{ \mathbf{x} \in \mathbb{F}^n : \mathbf{A}\mathbf{x} = \lambda\mathbf{x} \}$.
- The number of times a particular eigenvalue appears as a root of the characteristic polynomial is the algebraic multiplicity of the eigenvalue.
- The dimension of the eigenspace of an eigenvalue is the geometric multiplicity of the eigenvalue.
- Geometric multiplicity $\leq$ Algebraic multiplicity for each distinct eigenvalue.
- The dimension of the eigenspace of an eigenvalue λ is the nullity of $(\mathbf{A} - \lambda\mathbf{I})$.

Symmetric Matrices

- Real Symmetric Matrix: All eigenvalues are real
- Real Symmetric Matrix: Eigenvectors of distinct eigenvalues are orthogonal

Theorems

- A matrix $\mathbf{A} \in \mathbb{F}^{n \times n}$ is said to be *diagonalizable* if there exists an invertible matrix $\mathbf{P} \in \mathbb{F}^{n \times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{F}^{n \times n}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$
- Theorem 1: A square matrix $\mathbf{P}$ is orthogonal if and only if $\mathbf{P}^T = \mathbf{P}^{-1}$.
- Theorem 2: Spectral Theorem: A real symmetric matrix is orthogonally diagonalizable.
- A square matrix is diagonalizable if and only if the geometric multiplicity of each eigenvalue equals the algebraic multiplicity.
- From Theorem 1 and Theorem 2, the geometric multiplicity of each eigenvalue equals the algebraic multiplicity for any real symmetric matrix.

Example

- $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$
- $\lambda = 1, 1, 4$
- For $\lambda = 1$, algebraic multiplicity: 2, geometric multiplicity: 2.
- Basis vectors for eigenspace: $\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{3/2}} \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix}$
- For $\lambda = 4$, algebraic multiplicity: 1, geometric multiplicity: 1.
- Basis vector for eigenspace: $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Construct a set of orthonormal eigenvectors for a $n \times n$ real symmetric matrix **A**

- The spectral theorem asserts that any real symmetric matrix **A** is diagonalizable.
- Consider a $n \times n$ real symmetric matrix **A**.
- The characteristic polynomial is an n^{th} order polynomial.
- An n^{th} order polynomial has n roots.
- The sum of algebraic multiplicities is also equal to n .
- Since the matrix **A** is diagonalizable, the geometric multiplicity equals the algebraic multiplicity for each eigenvalue.
- The sum of geometric multiplicities is therefore also equal to n .

Construct a set of orthonormal eigenvectors for a $n \times n$ real symmetric matrix $\mathbf{A}$

- Since the matrix $\mathbf{A}$ is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal to each other. In other words, the eigenspaces are mutually orthogonal.
- Construct a set of orthonormal eigenvectors for each eigenspace.
- The entire collection of eigenvectors is an orthonormal set.
- Since there are a total of n eigenvectors in $\mathbb{R}^n$, we have constructed an orthonormal basis for $\mathbb{R}^n$.
- We denote the set of eigenvectors as $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

Optimization Problem

$$\begin{array}{ll} \text{maximize} & \|\mathbf{Ax}\|^2 \\ \text{such that} & \|\mathbf{x}\| = 1 \end{array} \quad (1)$$

Optimization Problem

- A real symmetric matrix $A \in \mathbb{R}^{n \times n}$ is called *positive semidefinite* if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
- $(\mathbf{A}^T \mathbf{A})^T = \mathbf{A}^T (\mathbf{A}^T)^T = \mathbf{A}^T \mathbf{A}$
- $\mathbf{x}^T (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{x}^T \mathbf{A}^T)(\mathbf{A} \mathbf{x}) = (\mathbf{A} \mathbf{x})^T (\mathbf{A} \mathbf{x}) = \|\mathbf{A} \mathbf{x}\|^2 \geq 0$
- $\mathbf{A}^T \mathbf{A}$ is symmetric and positive semidefinite: Eigenvalues are non-negative
- Since $\mathbf{A}^T \mathbf{A}$ is symmetric, one can construct an orthonormal set of eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^n$:
 $\mathbf{A}^T \mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$
- $\sigma_i = \sqrt{\lambda_i}$ are the singular values of $\mathbf{A}$
- $\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$
- $\mathbf{v}_i^T \mathbf{v}_i = \|\mathbf{v}_i\|^2 = 1, \mathbf{v}_i^T \mathbf{v}_j = 0$
- $\mathbf{x}^T \mathbf{x} = 1 \Rightarrow c_1^2 + c_2^2 + \dots + c_n^2 = 1.$

Optimization Problem

$$\begin{aligned}\|\mathbf{Ax}\|^2 &= (\mathbf{Ax})^T \mathbf{Ax} = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} \\ \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} &= \mathbf{x}^T \mathbf{A}^T \mathbf{A}(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n) \\ &= \mathbf{x}^T (c_1 \mathbf{A}^T \mathbf{A} \mathbf{v}_1 + c_2 \mathbf{A}^T \mathbf{A} \mathbf{v}_2 + \dots + c_n \mathbf{A}^T \mathbf{A} \mathbf{v}_n) \\ &= \mathbf{x}^T (c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2 + \dots + c_n \lambda_n \mathbf{v}_n) \\ &= (c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n)^T (c_1 \lambda_1 \mathbf{v}_1 + \dots + c_n \lambda_n \mathbf{v}_n) \\ &= c_1^2 \lambda_1 + \dots + c_n^2 \lambda_n \\ &= c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2\end{aligned}\tag{2}$$

Optimization Problem

- Arrange singular values as $\sigma_1^2 \geq \sigma_2^2 \geq \dots \geq \sigma_n^2$
- $\|\mathbf{Ax}\|^2 = c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2 \leq c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_1^2 = \sigma_1^2 (c_1^2 + \dots + c_n^2) = \sigma_1^2$
- Select $\mathbf{x} = \mathbf{v}_1$
- $\mathbf{x} = \mathbf{v}_1 \Rightarrow c_1 = 1, c_2 = c_3 = \dots = c_n = 0$
- $\|\mathbf{Ax}\|^2 = c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2 = \sigma_1^2$
- The solution to the optimization problem is obtained by selecting $\mathbf{x} = \mathbf{v}_1$, i.e. the eigenvector corresponding to the largest eigenvalue.